



## **Predictive Placement & Skill Gap Analytics Platform**

Bridging the gap between Campus Preparation and Corporate Requirements.

## The Problem

- **Students** lack clarity on what exact skills are needed for their target roles.
- **Colleges** struggle with tracking the specific placement readiness of hundreds of students.
- **Recruiters** waste time sifting through resumes that don't precisely match their technical requirements.

## **The Solution**

**An intelligent, RBAC-driven platform featuring three dedicated portals:**

1.  **Student Portal**
2.  **College Portal**
3.  **Recruiter Portal**

Provides end-to-end recruitment lifecycle management and upskilling guidance.

## ✨ Unique Features

-  **Skill Radar & Gap Analysis:** AI-driven profiling of missing skills.
-  **Personalized Learning Paths:** Automated course recommendations.
-  **Predictive Candidate Match:** Match scores for recruiters based on hard skills and proficiencies.
-  **Drive Management:** Streamlined campus placement workflow.
-  **Interview Prep & Alumni Mentorship:** Built-in guidance ecosystem.

## **Technical Architecture**

### **Built on the scalable MERN Stack**

- **Frontend:** React.js (Vite), Tailwind CSS, Lucide Icons
- **Backend:** Node.js, Express.js (MVC Pattern)
- **Database:** MongoDB Atlas, Mongoose
- **Security:** JWT Authentication, Bcrypt Hashing
- **Architecture:** API-Driven, RESTful Principles

## MVC Backend Pattern

-  **Models:** Defining schemas ( `Student` , `Job` , `Candidate` )
-  **Controllers:** Business logic ( `Skill Gap Calculation` , `Auth` )
-  **Routes:** Exposing clean API endpoints

# Database Design

## Why NoSQL / MongoDB?

- Highly flexible schema for dynamic skill arrays and nested user profiles.
- Separation of concerns: the core `User` model handles authentication while referencing specialized profiles ( `Student` , `College` , `Recruiter` ).

## Conclusion

**SkillBridge** replaces manual, chaotic placement processes with a data-driven, predictive model.

It ensures **Students get skilled, Colleges look good, and Recruiters get the right talent securely.**



# Thank You!

Any Questions?